Network Topologies and VLANs

In any computer network, devices must be connected in some way so that they can communicate with one another. As networks grow larger, simply connecting devices is not enough. We also need to understand how data moves between devices and how a single physical network can be logically divided into multiple smaller networks.

This module focuses on three fundamental concepts:

Device Arrangement

Data Movement

Logical Division of Networks

Together, these concepts help us understand how modern Local Area Networks (LANs) are designed and managed.

Why Do We Need Network Topologies?

A Single Point of Failure (SPOF) is a component whose failure can cause an entire system or network to become unavailable. For example, if a network depends on a single device or communication path, the failure of that component can disrupt all communication.

To overcome this problem, networks often introduce redundancy by adding backup devices and alternate communication paths. However, once multiple devices and multiple paths exist, an important question arises:

How should all these devices be arranged within the network?

This arrangement of devices is one of the key ideas behind network topology.

No matter how many devices a network contains, they must be organized in a structured way to ensure efficient communication and reliability.

Two Fundamental Questions in Networking

Whenever a network is designed, two important questions must be answered:

How are the devices connected physically?

How does data flow logically between them?

Although these questions appear similar, they refer to two different aspects of network design.

Physical Topology

Physical topology describes how devices are connected in the real world. It focuses on the actual arrangement of devices, cables, switches, and links within a network.

In simple terms, physical topology answers questions such as:

What devices are connected together?

How are the cables arranged?

What does the network physically look like?

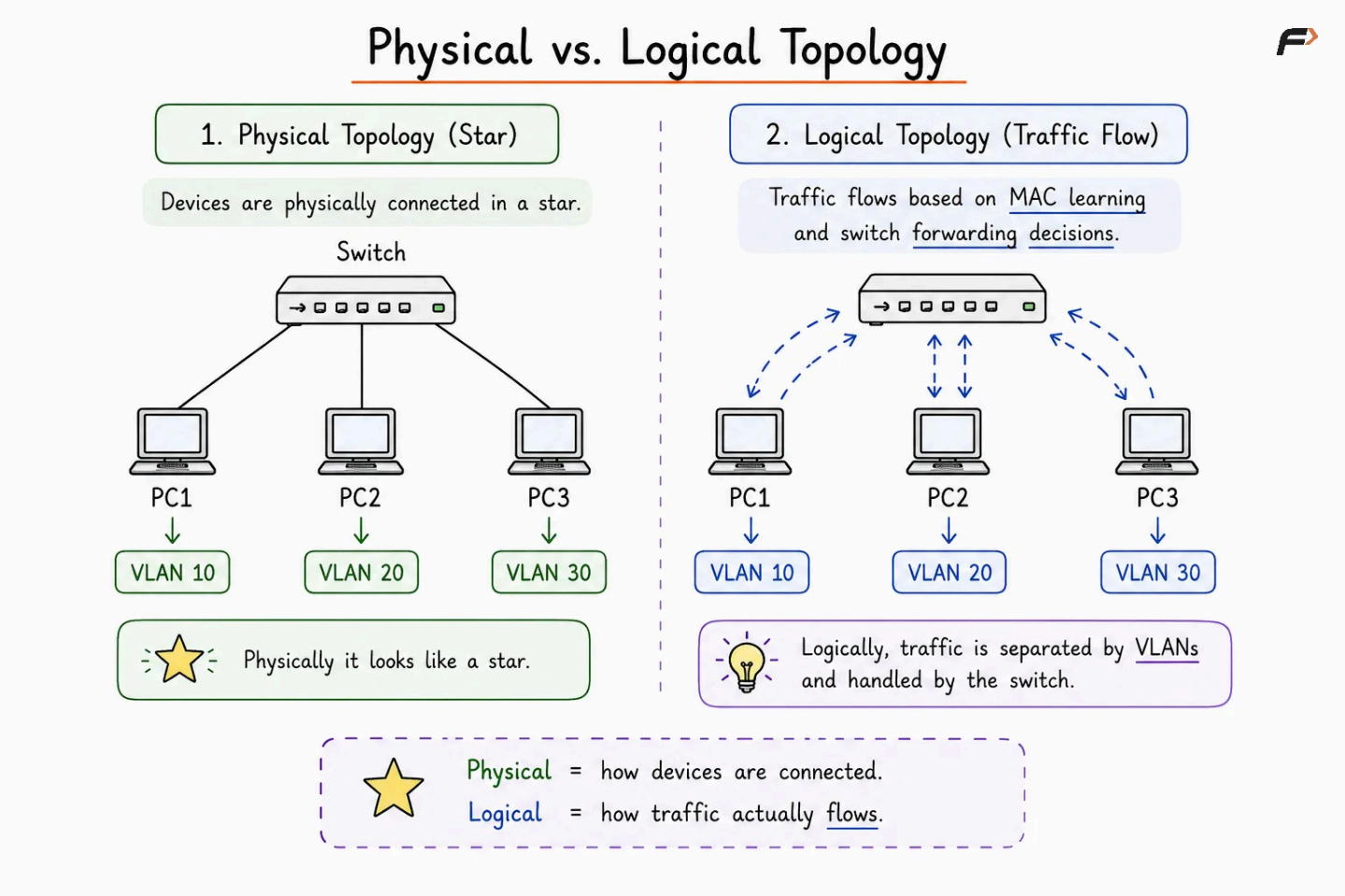
For example, multiple computers may be connected to a central switch. This arrangement represents the physical topology of the network.

A simplified representation would look like:

This diagram focuses only on how devices are physically connected.

Logical Topology

Logical topology describes how data actually moves and behaves within the network. It focuses on communication patterns rather than physical connections.



Logical topology answers questions such as:

Which devices can communicate directly?

How does traffic flow through the network?

What communication rules are applied?

Two devices may be physically connected to the same switch but still belong to different logical networks.

This means: Physical connectivity does not always determine logical communication.

Understanding VLANs

One of the most common ways to create logical separation inside a network is through a VLAN (Virtual Local Area Network).

A VLAN allows a single physical network to be divided into multiple logical networks without requiring separate switches or separate physical infrastructure.

Consider an office network. Physically, all employees may be connected to the same switch. However, the organization may want different departments to operate in separate logical

networks.

For example:

Engineering Department → VLAN 10

Sales Department → VLAN 20

Management Department → VLAN 30

Even though all devices are connected to the same physical switch, they are logically separated into different virtual networks.

VLAN Communication

When devices belong to the same VLAN, they can communicate directly with one another. However, devices belonging to different VLANs are logically separated.

For example:

PC1 belongs to VLAN 10

PC2 belongs to VLAN 20

PC3 belongs to VLAN 30

Even though all three computers may be connected to the same switch, they do not automatically communicate with one another because they belong to different logical networks.

To allow communication between VLANs, a routing device is required.

This routing function can be performed by:

A Router

A Layer 3 Switch

The router acts as the bridge that allows traffic to move from one VLAN to another.

Physical Topology vs Logical Topology

Physical Topology	Logical Topology
Describes how devices are connected.	Describes how data flows.
Focuses on cables, switches, and links.	Focuses on communication behavior.
Represents the actual network layout.	Represents communication rules and paths.
May remain unchanged.	Can change through VLANs and routing.

Key Takeaway

The most important concept introduced in this module is that physical arrangement and logical communication are not always the same thing.

A group of devices may share the same switch and physical infrastructure, yet belong to completely different logical networks through VLANs. In such cases, communication between those logical networks requires routing.

Understanding this distinction forms the foundation for learning network topologies, switch forwarding, MAC addresses, VLANs, and inter-VLAN routing in the upcoming lessons.

Physical Network Topologies

A physical topology describes the actual arrangement of devices, cables, links, and networking equipment in a network. It focuses on how devices are physically connected rather than how traffic logically behaves. Whenever a network is designed, one important question must be answered:

How should devices be connected together?

The answer to this question determines the network topology. Different topologies offer different trade-offs in terms of cost, scalability, fault tolerance, performance, and ease of management.

The major physical topologies are:

Star Topology

Ring Topology

Mesh Topology

Bus Topology

Tree Topology

Hybrid Topology

Why Topologies Matter

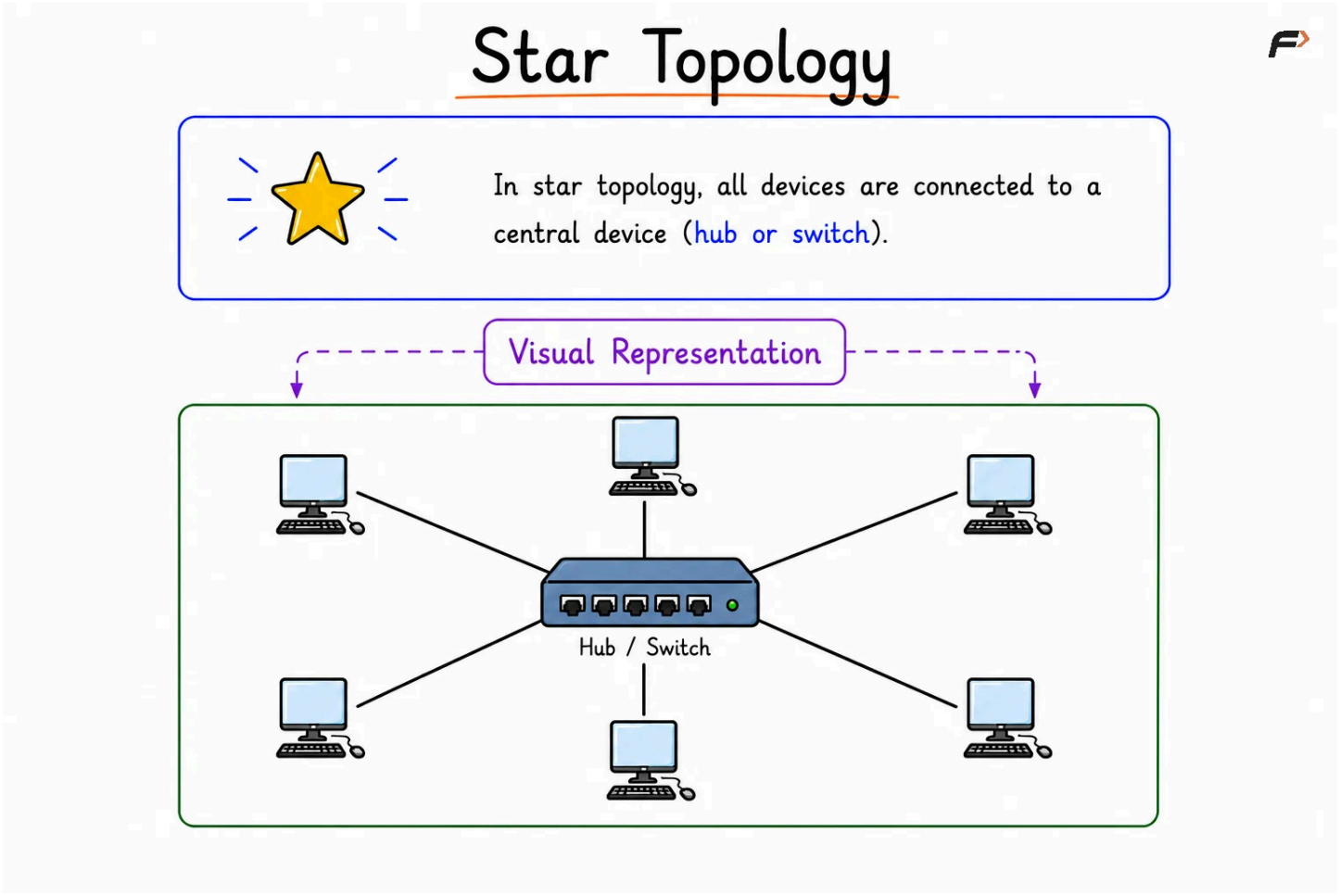
Earlier, we discussed the concept of a Single Point of Failure (SPOF). If a network depends entirely on a single device or communication path, failure of that component can disrupt communication for many users.

As networks grow larger, it becomes increasingly important to design them in a way that balances reliability, scalability, performance, and cost.

Network topologies provide structured ways to organize devices and communication paths. Some topologies prioritize simplicity and low cost, while others focus on redundancy and fault tolerance. Understanding these topologies helps network engineers choose the right design for different environments.

Star Topology

In a star topology, all end devices connect to a single central device. In modern networks, this central device is usually a switch. When we talk about a star topology, we are generally describing the local arrangement of devices within a LAN rather than the complete end-to-end path through the Internet.



For example, in a home network, the central device may be a Wi-Fi router with an integrated switch. In an office environment, the center is typically an Ethernet switch. All devices communicate through this central point.

Why Star Topology Is Common

A useful analogy is the airport hub model. Many flights connect through a major airport because it simplifies management and coordination. The same principle applies to networking. By having a central point, the network becomes easier to control and expand.

Star topology is widely used because it is:

Easier to manage

Easier to troubleshoot

Easier to scale

Well suited for Ethernet switching

How It Works

When one device sends data, the data first reaches the central switch. The switch then forwards the data to the appropriate destination device. All communication passes through

the central device.

Advantages

Easy to add new devices.

Failure of one device usually affects only that device.

Simple troubleshooting due to centralized connectivity.

Better performance with modern switches.

Works naturally with switched Ethernet networks.

Disadvantages

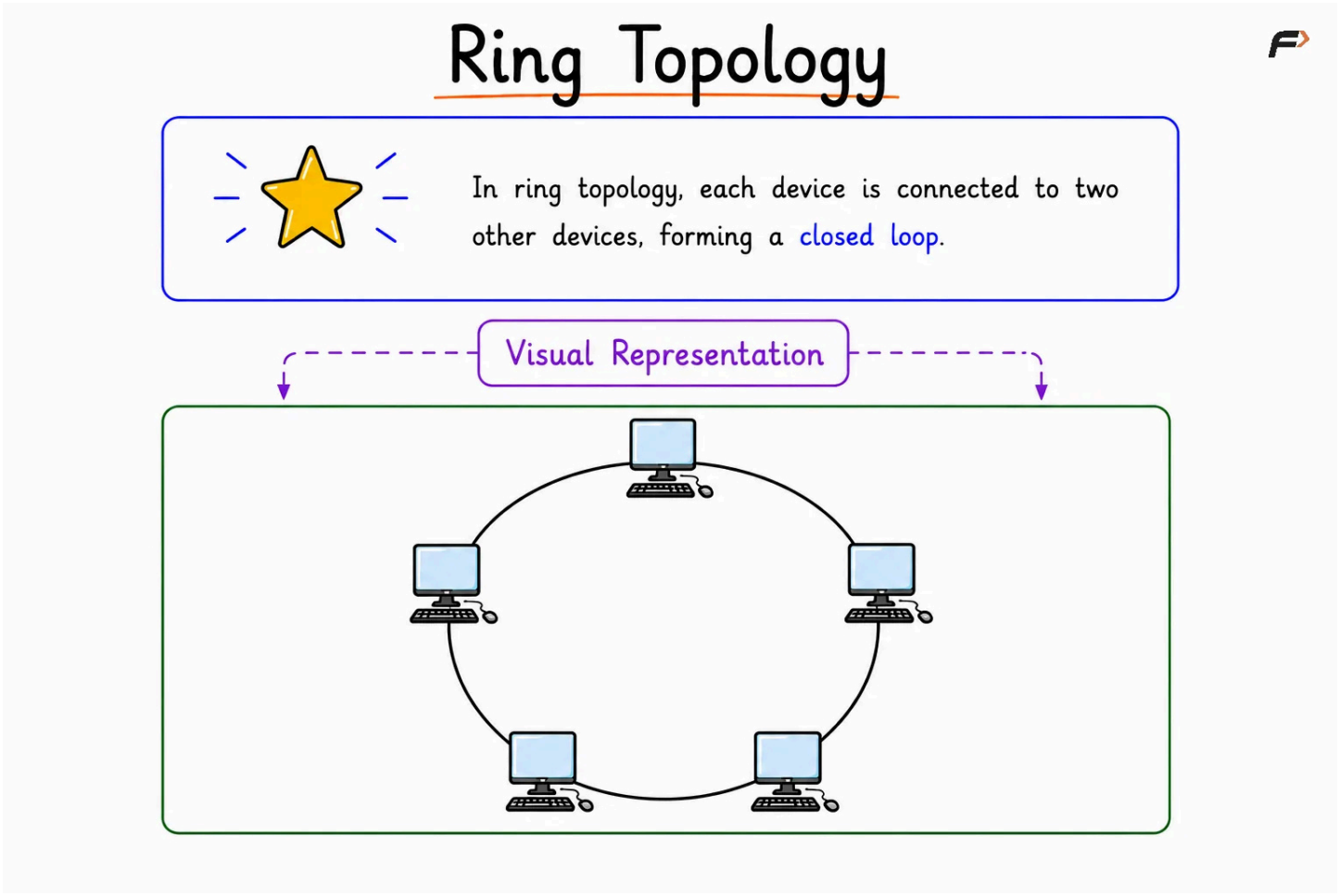
The central switch becomes a critical dependency.

If the switch fails, network connectivity may be disrupted.

Requires more cabling since every device needs a dedicated connection.

Ring Topology

In a ring topology, each device connects to two neighboring devices, forming a closed loop. A simple way to visualize this is to imagine people standing in a circle passing a note from one person to the next. The note continues moving around the circle until it reaches the intended recipient.



Why It Existed

Ring topology was useful in older networking systems because it allowed devices to access the network in a predictable and orderly manner. Rather than allowing multiple devices to compete for the medium simultaneously, communication could be controlled in sequence.

A classic example of this approach is **Token Ring**.

How It Works

When a device transmits data, the frame travels around the ring from one device to the next. Each intermediate device receives the frame, checks it, and forwards it until the destination is reached.

Token-Based Access

Some ring networks use a token. Only the device currently holding the token is allowed to transmit data. This prevents multiple devices from transmitting simultaneously and significantly reduces collisions.

Advantages

Predictable access in token-based systems.

Reduced collisions.

Useful in environments requiring deterministic communication.

Can support redundancy in modern implementations.

Disadvantages

A single break can disrupt communication if protection mechanisms are absent.

More difficult to manage than modern star networks.

Data may pass through multiple nodes, increasing delay.

Where Ring Concepts Are Still Used

Traditional Token Ring networks are now rare, but ring concepts still appear in:

Metro fiber networks

Industrial Ethernet deployments

Telecommunication transport networks

In these environments, ring designs are primarily used for resilience and fault tolerance rather than classic Token Ring behavior.

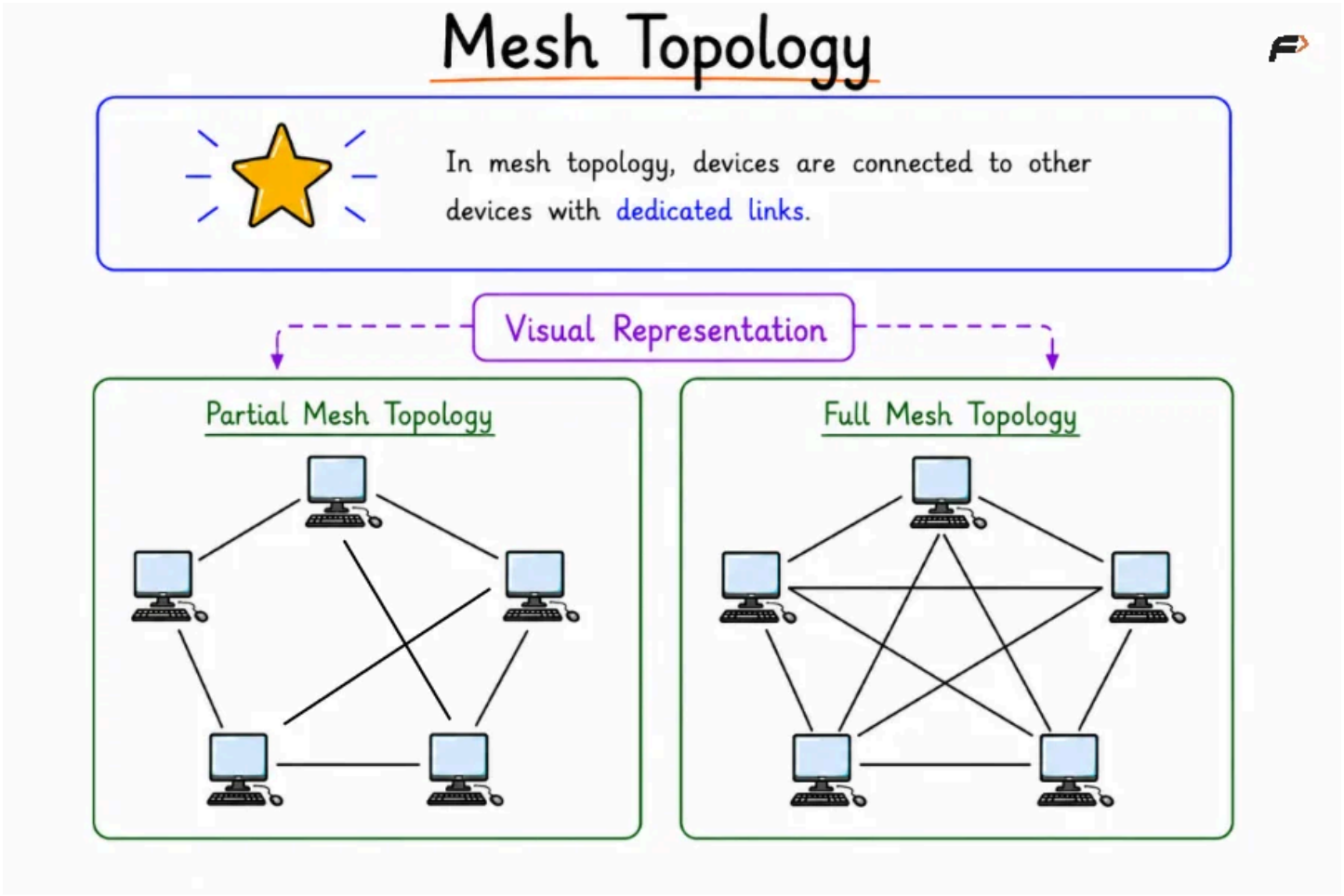
Mesh Topology

In a mesh topology, devices are connected to many or all other devices. The primary idea behind mesh networking is to provide multiple possible communication paths between nodes.

There are two common forms of mesh topology:

Full Mesh: Every device has a direct connection to every other device.

Partial Mesh: Only some devices have multiple direct connections.



Road Network Analogy

Mesh topology can be compared to a road network with multiple alternate routes. If one road becomes blocked, traffic can simply take another route and still reach its destination. The availability of multiple paths improves resilience and flexibility.

Why Mesh Topology Is Used

Redundancy

Fault tolerance

Multiple communication paths

Improved availability

Better path selection

Advantages

High redundancy.

Excellent fault tolerance.

Suitable for critical environments.

Supports alternate communication paths.

Disadvantages

Expensive to deploy.

More complex to manage.

Requires more cables and ports.

Full mesh becomes impractical as the network grows.

Full Mesh Scaling Problem

For a full mesh containing **n** devices: Links = n(n - 1) / 2

The number of required links grows rapidly as more devices are added. This is why full mesh topologies are rarely used at large scales.

Modern Data Center Networks

Modern data centers are usually not literal full meshes. Instead, they often use architectures inspired by mesh concepts, such as Spine-Leaf designs.

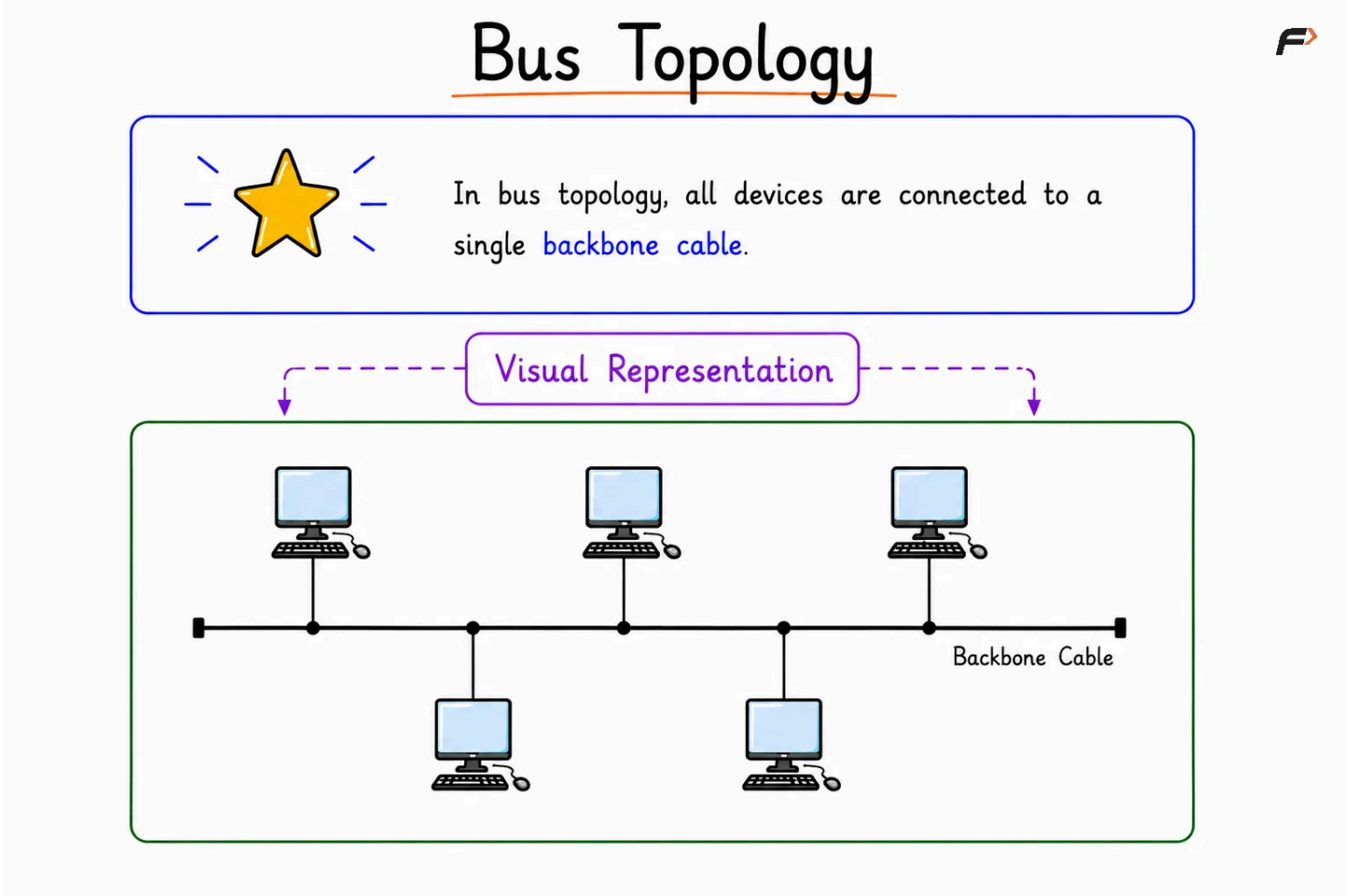
Leaf Switches: Connect servers and lower-layer networks.

Spine Switches: Form the high-speed core.

Multiple Route Options: Provide redundancy, load balancing, and scalability.

Bus Topology

In a bus topology, all devices share a single common backbone cable. A simple analogy is a shared road where every participant places messages onto the same communication path.



All communication uses the same shared medium.

Why It Existed

Early Ethernet networks used bus topology because it required less cabling than connecting every pair of devices individually. Instead of dedicated connections, all devices attached to a common backbone.

How It Works

A device places a signal onto the shared cable.

The signal propagates across the network.

All devices receive the transmission.

Only the intended recipient processes the data.

Other devices ignore the frame.

Collision Risk

Because all devices share the same medium, multiple devices may attempt to transmit simultaneously. This creates collisions. Early Ethernet networks addressed this using

CSMA/CD (Carrier Sense Multiple Access with Collision Detection).

Advantages

Lower infrastructure cost in older deployments.

Reduced cabling requirements.

Simple physical design.

Disadvantages

Single point of failure in the backbone cable.

Collisions increase as more devices are added.

Poor scalability.

Difficult troubleshooting.

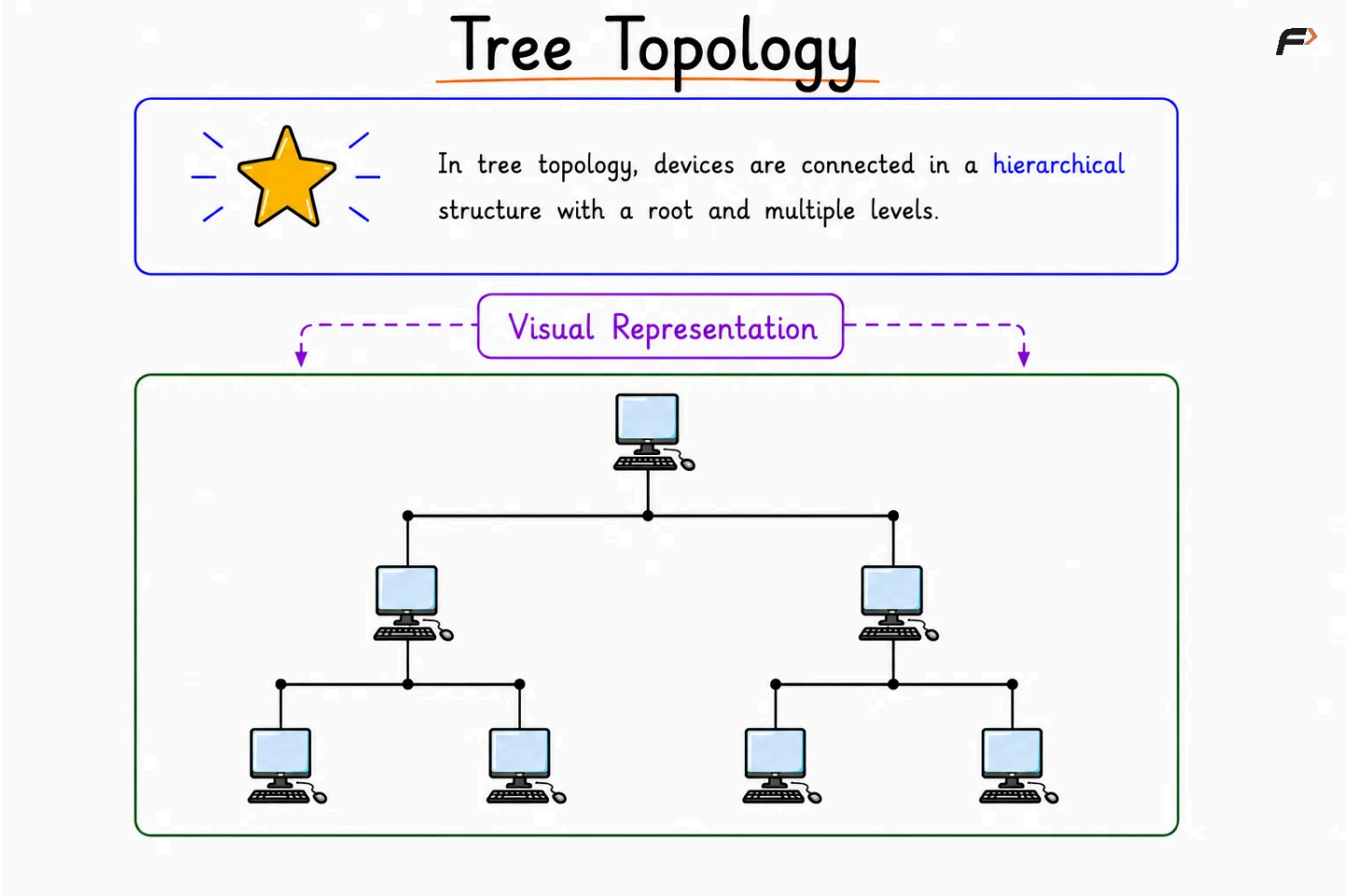
Performance decreases under heavy load.

Modern Relevance

Pure bus topology is largely historical. Modern switched Ethernet networks replaced shared-medium communication with dedicated point-to-point links connected through switches.

Tree Topology

Tree topology organizes devices in a hierarchical structure. It can be viewed as multiple star topologies connected across different levels. The resulting structure resembles a tree, with higher-level devices branching out to lower-level devices and eventually reaching end hosts.



Why It Exists

As networks grow, a single switch is often insufficient. Additional switches are needed to connect more users, rooms, floors, departments, or buildings. Tree topology provides a scalable way to organize these larger networks.

Tree topology offers:

Scalability

Structured design

Easier expansion

Better traffic organization

Network Layers in a Tree Topology

Core Layer

The high-speed backbone that interconnects major parts of the network and carries large volumes of traffic.

Distribution Layer

Acts as an aggregation and policy layer, often handling routing, filtering, and path selection.

Access Layer

The user-facing layer where end devices connect to the network.

Advantages

Highly scalable.

Easy to expand incrementally.

Structured and organized.

Well suited for enterprise environments.

Disadvantages

Higher-level devices become critical dependency points.

More design complexity.

Requires additional planning and cabling.

Tree topology is often best understood as a hierarchy of star topologies. It is widely used in enterprise networks, corporate offices, and campus environments.

Conclusion

Network topology describes how devices and communication paths are arranged within a network. Different topologies are designed to solve different problems. Star topology dominates modern LANs because of its simplicity, ring topology provides predictable communication, mesh topology emphasizes redundancy, bus topology represents an important historical design, tree topology enables large-scale hierarchical networks, and hybrid topology combines multiple approaches to meet real-world requirements.

Understanding these topologies is essential because they influence network performance, scalability, reliability, fault tolerance, and overall design decisions in modern computer networks.

Hybrid Topology

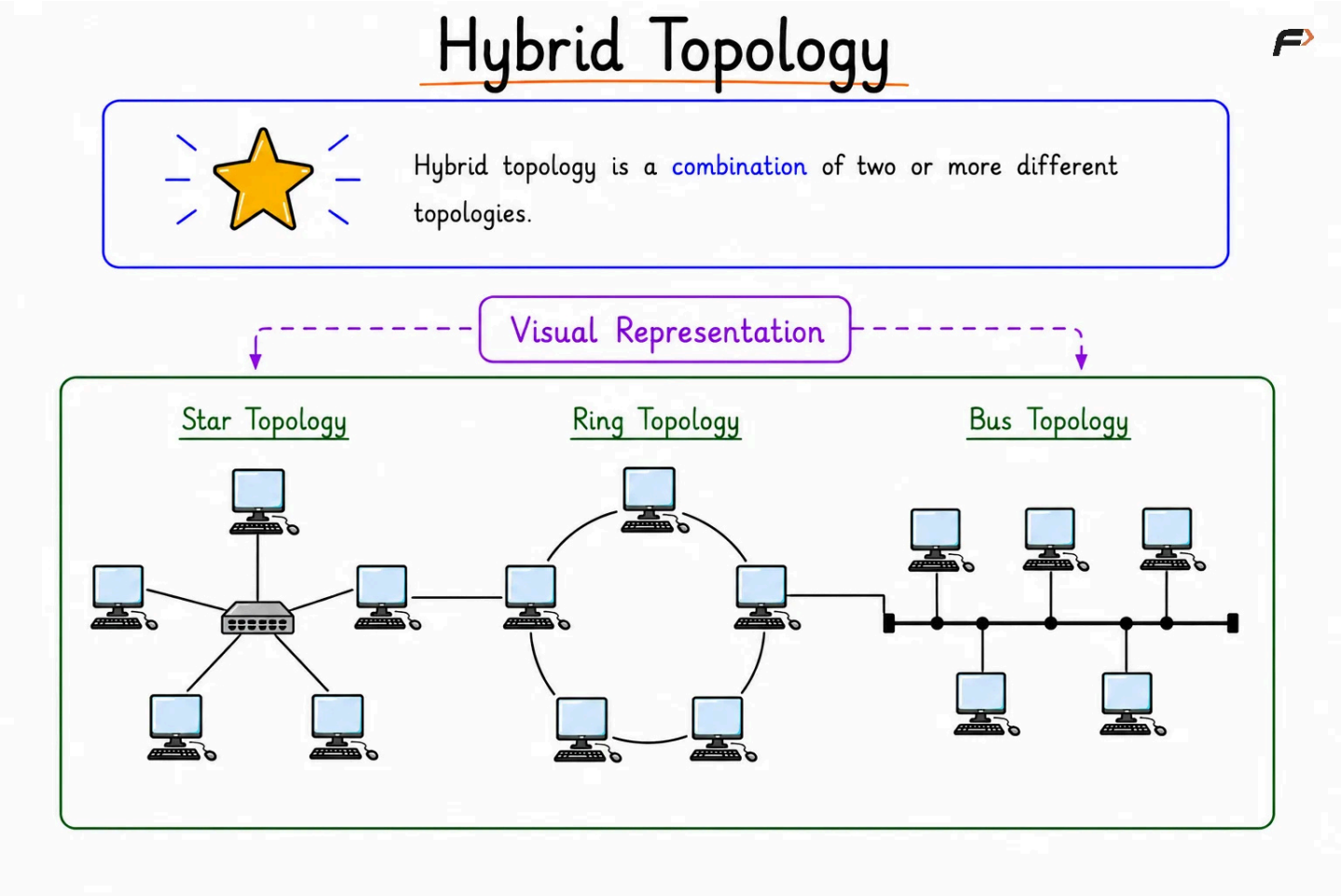
The name itself gives away the basic idea behind this topology. The word **Hybrid** refers to a combination of two or more different things. Similarly, a Hybrid Topology is a network

topology that combines two or more different network topologies within the same network infrastructure.

Instead of building an entire network using only a Star Topology, Mesh Topology, Bus Topology, or Tree Topology, different parts of the network can use different designs depending on their specific requirements.

What is a Hybrid Topology?

A Hybrid Topology is a network that combines two or more different physical topologies. It allows network designers to use the strengths of multiple topologies within a single network rather than relying on one design everywhere.



The key idea is simple:

Different parts of a real-world network often have different requirements, so they may need different topologies.

Understanding Hybrid Topology Through an Example

Consider an organization that has two major components:

An employee network where users work on laptops and desktops.

A data center containing critical servers and applications.

The requirements of these two environments are very different.

For employees, the organization may prioritize:

Low cost

Easy expansion

Simple management

In this case, a **Star Topology** is a good choice because all devices can communicate through a central switch while keeping the design simple and cost-effective.

The data center, however, may have completely different requirements. Since critical applications and services run inside the data center, the organization may require:

High availability

Redundancy

Multiple communication paths

Fault tolerance

To achieve these goals, a **Mesh Topology** may be used where devices have multiple connections to one another.

As a result:

Employee Network → Star Topology

Data Center → Mesh Topology

Since two different topologies are being used within the same organization, the resulting design is called a **Hybrid Topology**.

Why Do We Need Hybrid Topology?

Using the same topology everywhere is often impractical. For example, deploying a full mesh topology across an entire office would be extremely expensive and difficult to manage.

Similarly, using only a star topology in critical environments may not provide the redundancy required for high availability.

Hybrid topology solves this problem by allowing each part of the network to use the topology that best fits its requirements.

In simple terms:

One part of the network may prioritize simplicity.

Another part may prioritize redundancy.

A third part may prioritize scalability.

Hybrid topology allows all of these requirements to coexist within a single network.

Advantages of Hybrid Topology

Hybrid topology is widely used because it combines the strengths of multiple network designs.

Realistic for Real-World Systems: Most modern networks naturally contain multiple topology types.

Flexible Design: Different topologies can be combined based on business requirements.

Combines Simplicity and Redundancy: Some parts of the network can remain simple while others can be highly resilient.

Optimized for Cost and Performance: Expensive topologies can be used only where necessary.

Suitable for Large Networks: Hybrid designs scale well as organizations grow.

Supports Evolving Infrastructure: New network segments can be added without redesigning the entire network.

Disadvantages of Hybrid Topology

Although hybrid topology is highly flexible, it also introduces additional complexity.

More Complex to Design: Multiple topologies must be planned and integrated properly.

Harder to Understand: Different parts of the network may follow different communication patterns.

Troubleshooting Can Be Difficult: Problems may need to be investigated across multiple topology types.

Documentation Becomes Critical: Accurate network diagrams and records are essential.

Different Parts Behave Differently: Performance characteristics can vary across the network.

Why Hybrid Topology Is So Common

In practice, most networks are not built using a single topology. A full mesh topology everywhere would be prohibitively expensive, while a star topology everywhere may not provide the redundancy required for critical systems.

This is why modern enterprise networks, university campuses, cloud infrastructures, and data centers often combine multiple topology types. Different sections of the network are optimized for different goals, resulting in a hybrid design.

For this reason, Hybrid Topology is considered one of the most practical and commonly used network topologies in real-world deployments.

Key Takeaway

A Hybrid Topology combines two or more different network topologies within the same network. It allows organizations to balance simplicity, scalability, performance, cost, and redundancy by applying the most suitable topology to each part of the infrastructure. Because real-world networks have diverse requirements, hybrid topologies are far more common than any single pure topology.

Logical Topologies

When we started this module, we discussed three major ideas:

Device Arrangement

Data Movement

Logical Division of Networks

Device arrangement was covered through physical topologies, where we learned how devices, cables, switches, and links are physically connected. Data movement was covered in earlier lessons where we explored how information travels across a network.

The final piece of the puzzle is understanding how a single physical network can be logically divided into multiple networks and how traffic actually behaves within those networks. This concept is known as Logical Topology.

What is a Logical Topology?

A logical topology describes how data actually flows through a network regardless of the physical topology used to build that network. While physical topology focuses on the arrangement of devices and cables, logical topology focuses on communication behavior and traffic flow.

In simple terms:

Physical Topology tells us how devices are connected.

Logical Topology tells us how data behaves and moves.

A network may have one physical layout but exhibit a completely different logical behavior depending on how switching, forwarding, segmentation, and policies are configured.

Understanding Logical Behavior Through a Switch

Consider a simple star topology where multiple devices are connected to a switch. Physically, the network looks straightforward because every device connects to the central switch.

However, logical behavior depends on how the switch handles traffic.

Suppose device A wants to send data to device D for the first time. At this moment, the switch does not yet know where device D is located because its MAC address table is empty.

As a result, the switch performs a broadcast-like flooding operation and forwards the frame to all relevant ports.

This behavior can be represented as:

Device A sends data.

The switch receives the frame.

The MAC table does not contain the destination entry.

The switch forwards the frame to multiple ports.

This traffic behavior is part of the network's logical topology.

How MAC Learning Changes Traffic Flow

After the first communication, the switch begins learning. It records MAC addresses and the ports on which those devices are located. This information is stored in the MAC address table.

When device A sends data to device D again, the behavior changes.

The switch already knows where device D is located.

The MAC table contains the required entry.

The frame is forwarded directly to the destination port.

Flooding is no longer required.

Notice that the physical cables have not changed. The physical topology remains exactly the same. However, the behavior of traffic has changed because the switch has learned additional information.

This difference between physical connectivity and traffic behavior is one of the core ideas behind logical topology.

Logical Division of Networks

Logical topology also includes the concept of dividing a single physical network into multiple virtual networks.

Consider a switch with multiple ports. Instead of treating every port as part of the same network, we can logically separate them into different virtual networks.

For example:

Ports 1–3 → VLAN 10

Ports 4–6 → VLAN 20

Physically, all devices are connected to the same switch. However, logically they belong to different virtual networks. This process is known as logical network division.

The result is that one physical switch can behave as if it were supporting multiple independent networks.

One Switch, Multiple Virtual Networks

One of the most important ideas introduced in this lecture is: One physical switch can support multiple virtual networks.

For example, an organization may decide:

VLAN 10 → Engineering Team

VLAN 20 → Sales Team

Even though all employees are connected to the same physical switch, their communication behavior can be different because they belong to different logical networks.

As virtual networks are introduced, the switch must maintain additional information beyond a simple MAC-to-port mapping. This additional intelligence allows traffic to be forwarded more efficiently and according to logical boundaries.

Factors That Influence Logical Topology

Logical behavior is not determined solely by cables or physical connections. Several factors influence how traffic actually moves through a network.

1. Switching Behavior
2. Switching behavior determines how frames are forwarded. If the switch does not know the destination MAC address, it may flood the frame. If the destination is known, it forwards the frame directly to the appropriate port.
3. MAC Learning
4. Switches continuously learn and update MAC address information. Since devices may move, ports may change, and entries may expire, the MAC table is constantly evolving. These updates directly influence traffic flow.
5. VLAN Membership
6. Ports can belong to different virtual networks. Traffic forwarding decisions depend on which VLAN a device belongs to and whether communication is permitted across VLAN boundaries.

For example:

Engineering → VLAN 10

Sales → VLAN 20

If an employee moves from one department to another, the VLAN membership may also change, affecting logical communication behavior.

4. Broadcast Domains
4. Broadcast traffic does not always reach every device in the network. Broadcast behavior depends on logical boundaries.

For example, a broadcast generated within VLAN 10 may remain inside VLAN 10 rather than reaching devices in VLAN 20.

Therefore, logical topology influences who receives broadcast traffic.

5. Routing Boundaries
5. Communication within a network may be switched locally, while communication between different networks may require routing.

The point at which routing becomes necessary also influences how traffic flows through the network.

6. Policies and Access Control
6. Organizations often implement policies that control who can communicate with whom.

Examples include:

Firewall rules

Access Control Lists (ACLs)

Network Access Control policies

Security restrictions

Even if two devices are physically connected, these policies may allow, deny, or redirect communication, thereby affecting the logical topology.

Key Takeaway

Logical topology is concerned with how traffic behaves rather than how cables are arranged. Two networks with identical physical layouts may behave very differently because of MAC learning, switching decisions, VLAN configurations, routing boundaries, broadcast domains, and security policies.

Understanding logical topology is essential because modern networks are not defined solely by their physical structure. The way traffic is allowed to flow ultimately determines how the network operates.

Ethernet Switching

Ethernet switching is one of the most fundamental concepts in Local Area Networks (LANs). Most modern LANs rely on switches to move Ethernet frames between devices efficiently. While we have already seen how switches forward traffic at a high level, understanding what happens internally becomes even more important when virtual networks (VLANs) are introduced.

A switch operates primarily at the Data Link Layer (Layer 2) of the OSI model. Since it works with MAC addresses, its responsibility is to forward Ethernet frames from one device to another within a network.

Switch vs Hub

Before understanding Ethernet switching in detail, it is useful to compare a switch with a hub.

A hub is a simple device that acts as a blind repeater. Whenever it receives a frame, it sends that frame to every connected port without making any forwarding decisions.

A switch is much more intelligent. Instead of forwarding everything everywhere, it learns where devices are connected and forwards traffic only where it is needed.

Hub Switch

Repeats frames to all ports. Forwards frames selectively.

No learning mechanism. Learns MAC-to-port mappings.

Always floods traffic. Floods only when necessary.

Less efficient. More efficient.

This ability to learn and make forwarding decisions is what makes a switch significantly more efficient than a hub.

The Three Core Functions of a Switch

The behavior of a switch can be summarized through three primary actions:

1. Learn
2. Whenever a frame arrives, the switch observes the source MAC address. Over time, it learns which MAC addresses are reachable through which ports and stores this information in a MAC address table.
3. Forward
4. If the destination MAC address is already known, the switch forwards the frame only to the correct destination port instead of sending it everywhere.
5. Flood When Necessary
6. If the destination MAC address is unknown, the switch temporarily floods the frame so that the destination device can receive it and respond. Once the switch learns the destination's location, future communication becomes more efficient.

MAC Learning in Action

Consider a switch connected to several devices. Initially, the switch's MAC table is empty because it has no knowledge of which device is connected to which port.

Suppose Machine A sends a frame to Machine B. Since the switch does not yet know where Machine B is located, it floods the frame to other relevant ports. When Machine B responds, the switch observes the MAC addresses involved and updates its MAC table.

Over time, the table fills with MAC-to-port mappings. Once the switch knows where Machine B is connected, future frames destined for Machine B are forwarded directly instead of being flooded across the network.

This learning process is what makes Ethernet switching efficient.

The MAC Address Table (Forwarding Table)

A switch stores learned information in a structure known as the MAC Address Table, also called the Forwarding Table.

Traditionally, a forwarding table maps a MAC address to a switch port. However, modern switched networks often include VLANs, which require additional information to be stored.

A forwarding table typically contains:

MAC Address – The learned hardware address of a device.

VLAN – The virtual network to which the device belongs.

Port – The switch port through which the device is reachable.

Age Timer – Information used to determine when an entry should expire.

Why VLAN Information Is Stored

When virtual networks are introduced, MAC learning becomes VLAN-aware. The switch must not only know where a MAC address exists, but also the VLAN context in which it exists.

This ensures that forwarding decisions are made within the correct virtual network and prevents traffic from one VLAN from leaking into another VLAN.

Why Aging Matters

Devices do not remain connected to the same switch port forever. A laptop may disconnect from one port and reconnect to another, or a device may leave the network completely.

If MAC entries never expired, the switch could continue relying on outdated information. To avoid this problem, MAC table entries are associated with an age timer.

When an entry remains unused for a certain period, it is removed from the table. The next time traffic appears from that device, the switch relearns its location.

This keeps the forwarding table accurate and up to date.

Forwarding vs Flooding

The switch's behavior depends on whether it knows the destination MAC address.

Case 1: Known Unicast

If the destination MAC address already exists in the forwarding table, the switch forwards the frame only to the corresponding port.

For example, if the switch knows that Laptop B is connected to Port 5, any frame destined for Laptop B is forwarded directly to Port 5.

This is the most efficient form of switching and represents normal day-to-day traffic in a learned network.

Case 2: Unknown Unicast

If the destination MAC address is not present in the forwarding table, the switch cannot determine where the destination device is located.

In this case, it floods the frame to all relevant ports except the incoming port. Once the destination responds, the switch learns its location and updates the forwarding table.

Case 3: Broadcast Traffic

Sometimes a frame is intended for every device within a network rather than a single destination. Such traffic is called broadcast traffic.

When a broadcast frame arrives, the switch floods it to all ports that belong to the same VLAN, excluding the incoming port.

A common example is an ARP request, where a device asks:

Who has this IP address?

Since the destination is initially unknown, every device in the VLAN receives the request and the correct device replies.

Broadcasts and VLAN Boundaries

An important concept introduced in VLAN-aware switching is that broadcasts remain inside their VLAN.

For example:

Ports 1–6 belong to the Engineering VLAN.

Ports 7–12 belong to the Sales VLAN.

If a device in the Engineering VLAN sends a broadcast frame, the switch forwards it only to other Engineering VLAN ports. Devices in the Sales VLAN do not receive the broadcast.

This logical separation helps reduce unnecessary traffic and keeps different virtual networks isolated from one another.

Common Problems in Ethernet Switching

Although switching is highly efficient, certain problems can still occur within a network.

Unknown Destination Flooding

If the switch frequently encounters destination MAC addresses that are not present in its forwarding table, it may repeatedly flood traffic. Excessive flooding increases unnecessary network traffic and reduces efficiency.

Broadcast Storms

Large amounts of broadcast traffic can consume bandwidth and overwhelm devices. When broadcasts become excessive, normal communication may be severely affected.

Switching Loops

A loop occurs when frames continuously circulate through the network. Loops can generate duplicate frames, unstable MAC tables, and broadcast storms.

This is one of the primary reasons protocols such as Spanning Tree Protocol (STP) exist.

MAC Flapping

MAC flapping occurs when the same MAC address repeatedly appears on different switch ports. This often indicates a loop, misconfiguration, or unexpected device movement.

As a result, the switch may continuously update its forwarding table and exhibit unstable forwarding behavior.

Wrong VLAN Assignment

If a device is assigned to the wrong VLAN, communication may fail because the device is placed in the wrong logical network.

Incorrect VLAN assignments can create both connectivity and security issues.

Key Takeaways

A switch operates primarily at Layer 2 and forwards Ethernet frames using MAC addresses.

Unlike a hub, a switch learns MAC-to-port mappings and forwards traffic intelligently.

The forwarding table stores MAC addresses, VLAN information, ports, and aging information.

Known destinations are forwarded directly, while unknown destinations are flooded.

Broadcast traffic is flooded only within the appropriate VLAN.

VLAN-aware switching allows one physical switch to support multiple logical networks.

Common switching challenges include flooding, broadcast storms, loops, MAC flapping, and incorrect VLAN assignments.

Virtual Local Area Networks (VLANs)

A VLAN stands for **Virtual Local Area Network**. The word *virtual* is the most important part of the definition because VLANs create a logical separation rather than a physical one.

A VLAN is a logical way to split a single physical switched network into multiple separate Layer 2 networks. Even though devices may be connected to the same physical switch, VLANs make them behave as if they belong to different independent LANs.

From a physical perspective, all devices may still be connected to the same switch infrastructure. However, from a logical perspective, the switch behaves as if it contains multiple separate networks.

Why Do VLANs Exist?

To understand why VLANs are needed, consider a large switched LAN without any VLAN configuration.

In such a design, the entire network becomes one large broadcast domain. Whenever a switch needs to flood traffic, that traffic can potentially reach a very large number of devices.

As the network grows, this creates several problems:

Too Many Broadcasts: Excessive broadcast traffic consumes bandwidth and reduces efficiency.

Poor Isolation: All devices remain part of the same logical network.

Security Concerns: Unnecessary exposure exists when unrelated devices share the same broadcast domain.

Management Difficulties: Organizing users, policies, and departments becomes harder.

No Logical Separation: Different groups of users are forced into the same network.

These issues become increasingly significant as the number of devices connected to the network grows.

The Core Idea Behind VLANs

VLANs solve these problems by creating multiple independent broadcast domains inside a single switch.

For example, a switch may be configured as:

Ports 1–10 → VLAN 10

Ports 11–20 → VLAN 20

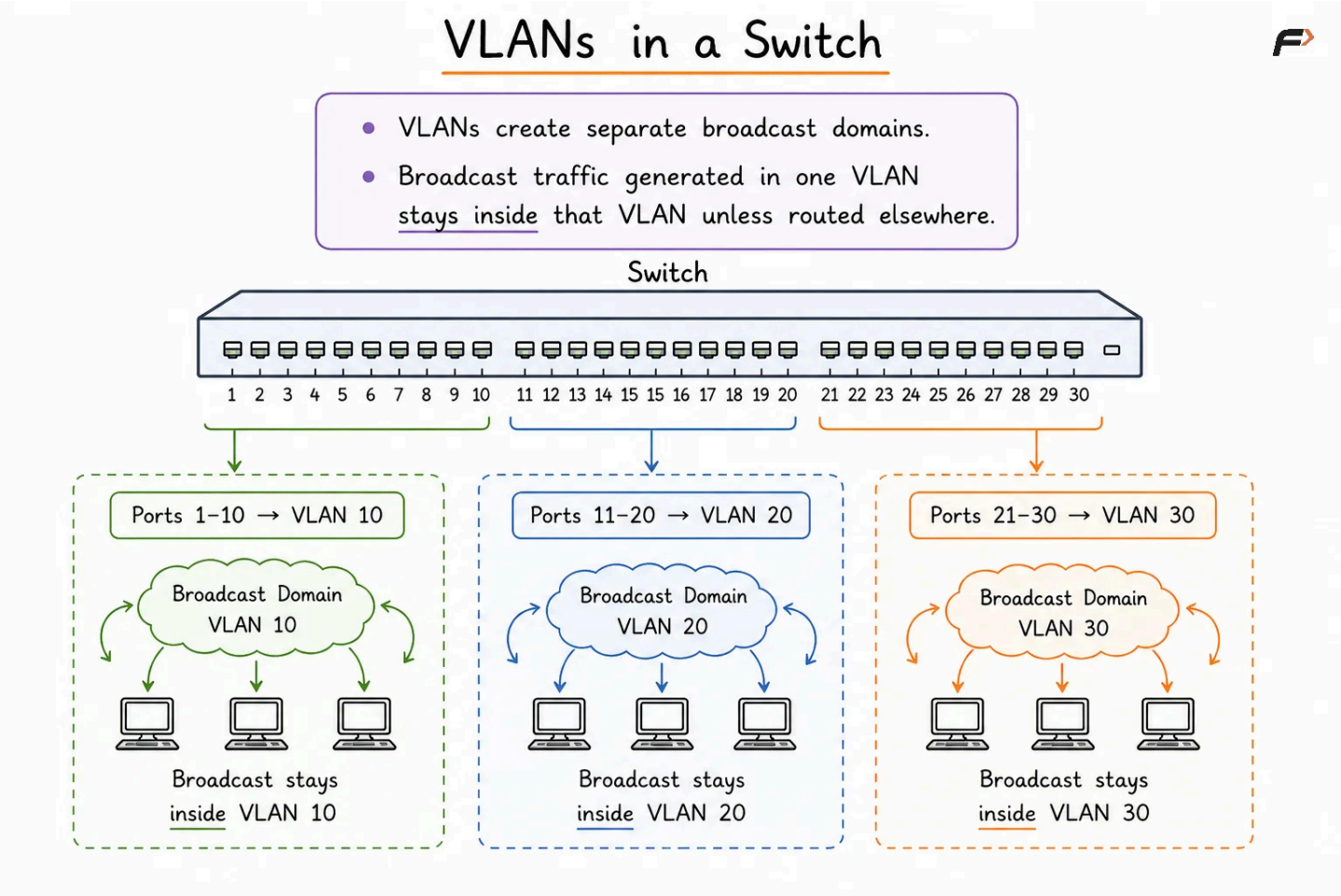
Ports 21–30 → VLAN 30

These numbers are called **VLAN IDs**. They simply identify different virtual networks. The specific numbers themselves are not special; they are identifiers used to distinguish one VLAN from another.

The important point is that traffic generated in one VLAN remains within that VLAN unless routing is introduced.

For example, if a device in VLAN 10 sends a broadcast frame, that broadcast remains inside VLAN 10. Devices in VLAN 20 and VLAN 30 do not receive it.

As a result, a single physical switch can behave like multiple separate Layer 2 networks.



Physical Network vs Logical Network

One of the key ideas emphasized in VLANs is that the physical network and the logical network are not always the same thing.

Physically, all devices may be connected to the same switch. If you simply look at the cables, it may appear to be one large network.

Logically, however, the switch can divide those devices into separate virtual networks using VLAN configuration.

This is why VLANs are considered a logical separation rather than a physical separation.

VLAN Membership

VLAN membership determines how devices participate in virtual networks.

It answers three important questions:

Which VLAN does a port belong to?

Which logical LAN does a device belong to?

Which broadcast domain should the device's traffic remain inside?

Suppose Port 5 belongs to VLAN 10. Any device connected to Port 5 becomes part of VLAN 10 and participates in the VLAN 10 broadcast domain.

If that device sends a broadcast frame, the traffic remains within VLAN 10 rather than reaching devices in other VLANs.

Types of VLAN Ports

A switch port can generally be configured in two different ways:

Access Port

Trunk Port

Access Port

An access port carries traffic for only one VLAN. These ports are typically used to connect end devices such as:

Desktop Computers

Laptops

Printers

IP Phones

Since the connected device belongs to a single VLAN, all traffic on that access port is associated with that VLAN.

For example:

Port 2 belongs to VLAN 10.

A laptop is connected to Port 2.

The laptop automatically becomes part of VLAN 10.

The end device generally does not need to understand VLAN information. From the device's perspective, it is simply connected to a normal network.

Untagged Frames on Access Ports

Frames sent to end devices are typically untagged. The switch already knows which VLAN a port belongs to, so the VLAN information does not need to be carried all the way to the endpoint device.

When a frame arrives on an access port, the switch identifies the VLAN using the port configuration itself.

For example:

Frame arrives on Port 2.

Port 2 belongs to VLAN 10.

The switch automatically associates the frame with VLAN 10.

The end device does not need to interpret any VLAN tags.

Trunk Port

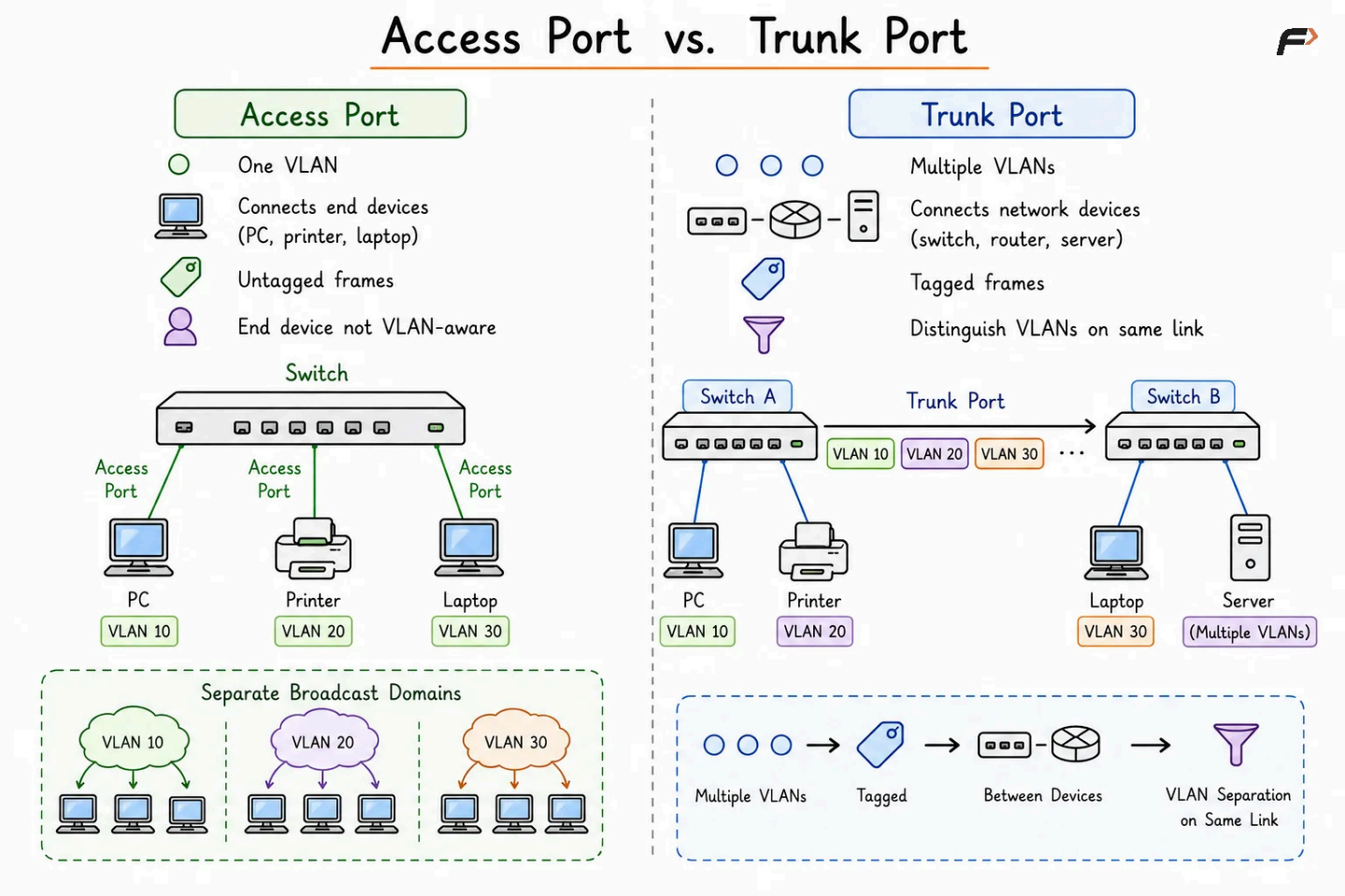
Unlike an access port, a trunk port carries traffic for multiple VLANs simultaneously.

Trunk ports are commonly used:

Between switches.

Between a switch and a router.

Between a switch and certain servers.



Because multiple VLANs share the same physical link, the receiving device must be able to distinguish traffic belonging to different VLANs.

This is where VLAN tagging becomes necessary.

Why Tagging Is Required

Imagine two switches connected through a single cable.

If VLAN 10, VLAN 20, and VLAN 30 traffic all travel through the same physical link, the receiving switch needs a way to identify which VLAN each frame belongs to.

Without VLAN identification:

Traffic from different VLANs would mix together.

The receiving switch would not know how to classify frames.

Broadcast boundaries would be lost.

Logical separation would break down.

To solve this problem, switches use VLAN tagging.

802.1Q VLAN Tagging

The most common VLAN tagging standard is **IEEE 802.1Q**.

With 802.1Q tagging:

A VLAN identifier is inserted into the Ethernet frame.

The frame carries information about its VLAN membership.

The receiving switch reads the tag.

The frame is placed into the correct VLAN.

This allows multiple VLANs to share a single physical trunk link while still remaining logically separated.

Common VLAN Use Cases

One of the most common uses of VLANs is departmental isolation.

For example:

HR Department → VLAN 10

Engineering Department → VLAN 20

Finance Department → VLAN 30

Even though all employees may use the same switching infrastructure, each department remains logically separated from the others.

Another common use case is guest Wi-Fi.

Organizations often place guest users into a separate VLAN so that visitors cannot directly access internal corporate devices and resources.

Can Devices in Different VLANs Communicate?

Not directly through Layer 2 switching.

Each VLAN is a separate broadcast domain and behaves like an independent Layer 2 network. Because of this separation, devices in different VLANs cannot simply exchange frames

through normal switching.

For example:

PC A belongs to VLAN 10.

PC B belongs to VLAN 20.

PC A cannot directly send Layer 2 frames to PC B through ordinary switching because both devices belong to different VLANs.

Inter-VLAN Routing

If communication is required between VLANs, a Layer 3 device must be involved.

This can be:

A Router

A Layer 3 Switch

The process of allowing communication between different VLANs is known as **Inter-VLAN Routing**.

The Layer 3 device receives traffic from one VLAN, routes it, and forwards it into the destination VLAN.

Key Takeaways

A VLAN is a logical separation of a physical switched network.

VLANs create multiple independent broadcast domains.

Ports can be assigned to different VLANs using VLAN IDs.

Access ports carry traffic for a single VLAN.

Trunk ports carry traffic for multiple VLANs.

802.1Q tagging preserves VLAN information across trunk links.

One physical switch can support multiple logical networks.

Devices in different VLANs require routing to communicate.

Switching vs Routing

Switching and routing are two fundamental concepts in computer networking. Both are responsible for moving data from one place to another, but they operate at different layers of the network and solve different communication problems.

A common way to understand the difference is:

Switching moves data within the same network.

Routing moves data between different networks.

Although both technologies help data reach its destination, they use different addressing schemes, different forwarding mechanisms, and different network devices.

What is Switching?

Switching is the process of forwarding data within the same local network. A switch receives Ethernet frames and decides which device should receive them based on MAC addresses.

For example, if two devices belong to the same VLAN or the same local network, a switch can forward traffic directly between them without involving a router.

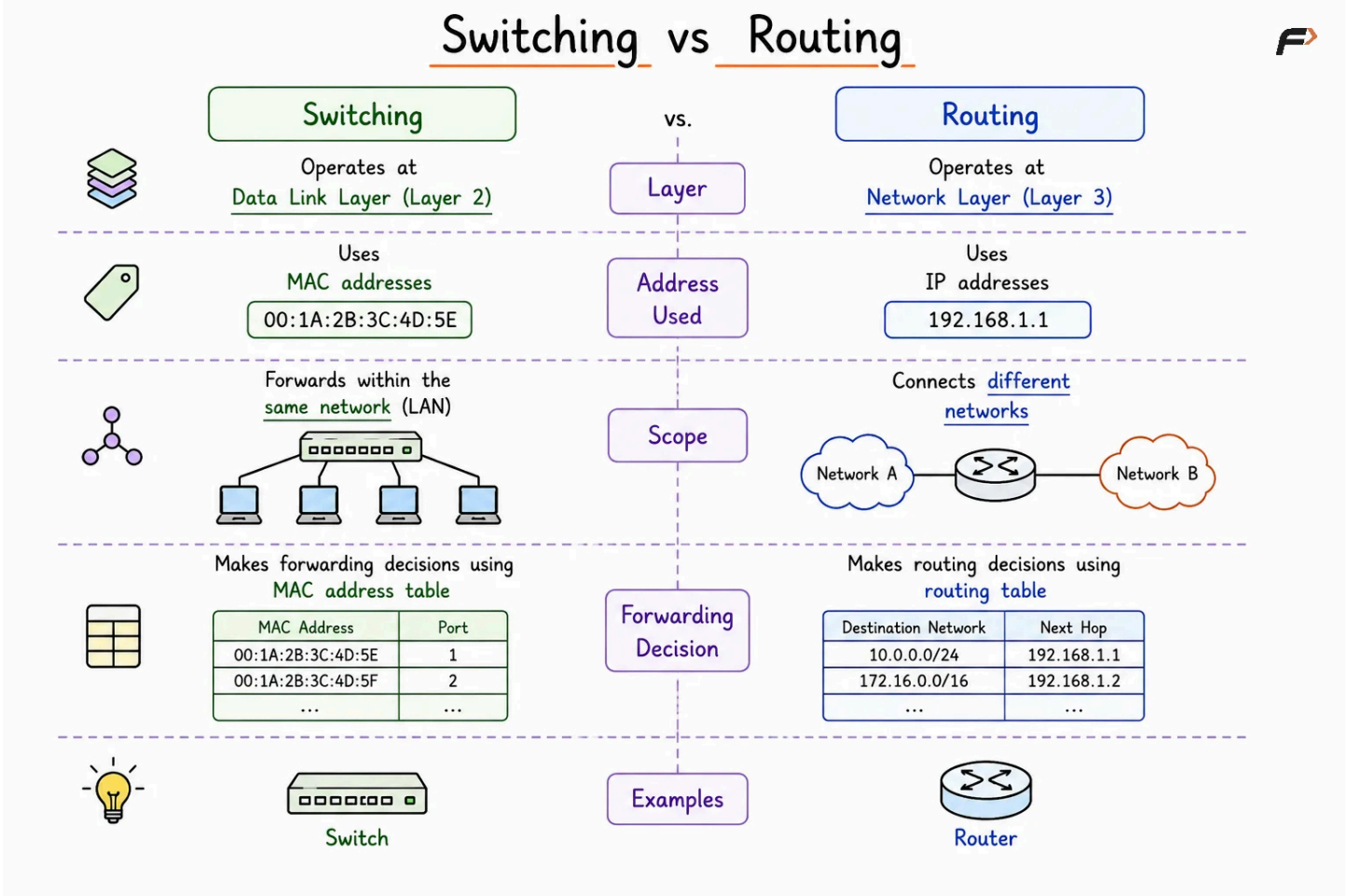
In simple terms: Switching is concerned with communication inside the same network.

What is Routing?

Routing is the process of moving data between different networks. When the source and destination belong to different networks, a router is required to determine how the data should reach its destination.

Instead of using MAC addresses to make decisions, routers use IP addresses and routing information to determine the best path for forwarding packets.

In simple terms: Routing is concerned with communication between different networks.



Layer 2 vs Layer 3

One of the most important differences between switching and routing is the OSI layer at which they operate.

Switching primarily operates at **Layer 2 (Data Link Layer)**.

Routing primarily operates at **Layer 3 (Network Layer)**.

Because switches operate at Layer 2, they work with MAC addresses and Ethernet frames. Routers operate at Layer 3 and work with IP addresses and packets.

MAC Addresses vs IP Addresses

Switches and routers make forwarding decisions using different types of addresses.

A switch uses the destination MAC address contained inside an Ethernet frame to determine where the frame should be forwarded.

A router uses the destination IP address contained inside an IP packet to determine which network the packet should travel toward.

Therefore:

Switching → MAC Address Based

Routing → IP Address Based

Frame vs Packet

Another important distinction is the type of data unit each technology handles.

In switching, the data unit is a **Frame**. An Ethernet frame contains information such as:

Source MAC Address

Destination MAC Address

Payload (IP Packet)

FCS and other Ethernet information

In routing, the data unit is a **Packet**. Routers examine packet-level information, particularly the destination IP address, to determine where the packet should be forwarded.

Therefore:

Switching → Frame

Routing → Packet

MAC Table vs Routing Table

Switches and routers maintain different types of forwarding information.

A switch maintains a **MAC Address Table** (also called a forwarding table). This table stores mappings between MAC addresses and switch ports.

A router maintains a **Routing Table**. This table helps determine where packets should be forwarded based on destination IP addresses and available network paths.

In both cases, the device maintains information that helps it make forwarding decisions efficiently.

Example of Switching

Consider a laptop and a printer connected to the same VLAN.

When the laptop sends data to the printer:

Both devices belong to the same network.

The switch uses MAC addresses to identify the destination.

The traffic remains within the local network.

No routing is required.

This is a typical example of switching.

Example of Routing

Now consider a laptop in VLAN 10 that needs to communicate with a server in VLAN 20.

Since VLAN 10 and VLAN 20 are different networks, ordinary Layer 2 switching is not sufficient.

The communication flow becomes:

Laptop sends traffic to the switch.

The switch forwards the traffic to a router or Layer 3 device.

The router performs routing.

The traffic is forwarded into VLAN 20.

The server receives the packet.

This is a typical example of routing.

Another Simple Way to Remember

Switching: Data stays inside the same local network.

Routing: Data travels to a different network.

This simple distinction is often enough to answer many interview questions about switching and routing.

Switching vs Routing Comparison

Aspect	Switching	Routing
Purpose	Moves data within the same network.	Moves data between different networks.
OSI Layer	Layer 2 (Data Link Layer)	Layer 3 (Network Layer)
Address Used	MAC Address	IP Address
Data Unit	Frame	Packet
Forwarding Information	MAC Address Table	Routing Table
Scope	Same VLAN or same local network.	Different VLANs or different networks.
Example	Laptop communicating with a printer in the same VLAN.	Laptop in VLAN 10 communicating with a server in VLAN 20.